

UBUNTU SERVER 26.04 LTS ENTERPRISE INSTALLATION GUIDE

ITEP 414 - System Administration and Maintenance
Week 3 Portfolio Project

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ITEP 414 - System Administration and Maintenance

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Scenario: ABC Startup Solutions

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1. Project Overview

Operating Systems serve as the foundation of every enterprise IT infrastructure. Before a server can host websites, databases, cloud services, or enterprise applications, it must first be properly installed, configured, secured, and documented.

In this project, I assumed the role of a **Junior System Administrator** for **ABC Startup Solutions** and deployed the company's first Linux server. The server will later be used for file sharing, remote administration, database hosting, web hosting, and internal services. This guide documents the complete installation of **Ubuntu Server 26.04 LTS** in a virtual machine so that a future administrator can reproduce the installation step by step.

2. Objectives

Type	Objectives
Knowledge	Explain the purpose of an operating system in enterprise environments; differentiate BIOS and UEFI firmware; explain the stages of the computer boot process; compare Ubuntu Server, Windows Server, and Rocky Linux.
Skills	Install Ubuntu Server in a virtual machine; configure server settings during installation; enable secure remote administration using SSH; verify server functionality; document installation procedures; produce professional technical documentation.
Professional Outcomes	Installed Ubuntu Server; configured a working Linux server; produced a deployment guide; created installation documentation; updated GitHub portfolio; published a LinkedIn learning reflection.

3. Software Requirements

- **Oracle VirtualBox** (virtualization platform)
- **Ubuntu Server 26.04 LTS ISO** (latest stable version)
- **Windows Server 2025 Evaluation ISO** (bring-home activity)

4. Virtual Machine Specifications

Component	Minimum Requirement	Configured
Name	Ubuntu-Server-Week03	Ubuntu-Server-Week03
Operating System	Ubuntu Server LTS	Ubuntu Server 26.04 LTS
RAM	4 GB	4 GB
CPU	2 Virtual Processors	2
Storage	40 GB (VDI/VMDK)	40 GB
Network	NAT (or Bridged)	NAT
Optical Drive	Ubuntu Server ISO	Ubuntu Server 26.04 LTS ISO

5. Step-by-Step Installation Procedure (Part A)

The installation was performed using the Subiquity text-based installer inside Oracle VirtualBox.

Step 1 - Language and Keyboard Layout

Selected **English** for the installer language and **English (US)** as the keyboard layout.

Step 2 - Network Configuration

Accepted **DHCP** as instructed by the module. The installer detected the interface **enp0s3** and assigned the IP address **10.0.2.15/24** through the NAT network. This address was recorded for later verification.

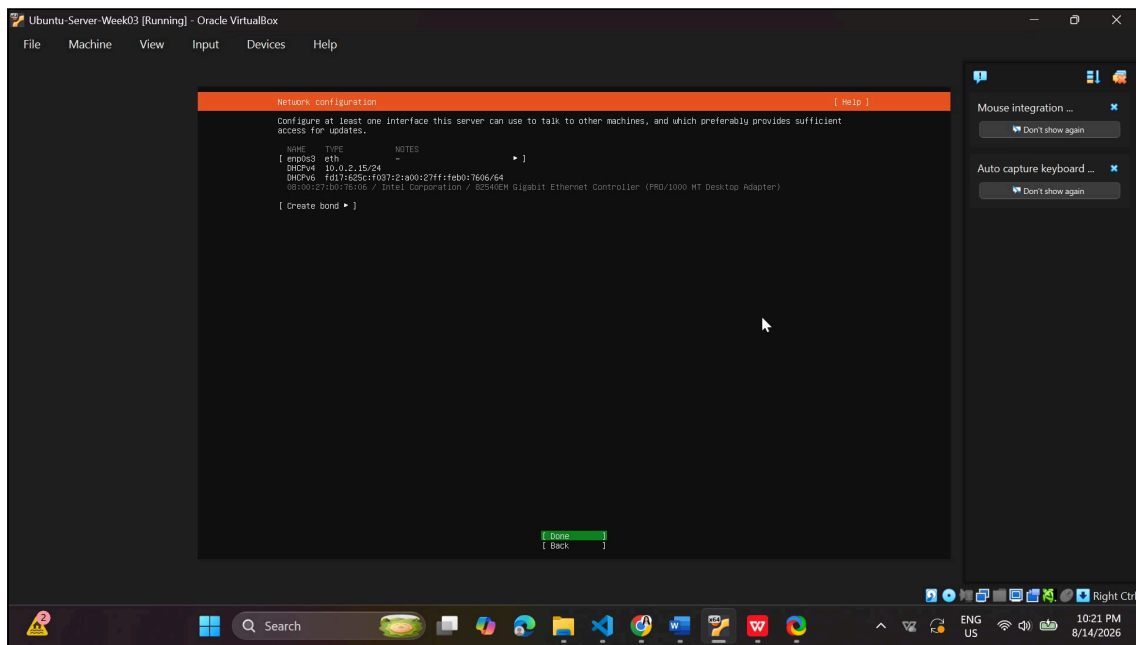


Figure 1. Network configuration showing interface enp0s3 with IP 10.0.2.15/24 (DHCP)

Step 3 - Hostname and User Account

Assigned the hostname **server01** as required by the module, and created a non-root administrative user:

Field	Value
Full Name	Mazel Catherine P. Joya
Server Name	server01
Username	cathy
Password	Strong password (hidden)

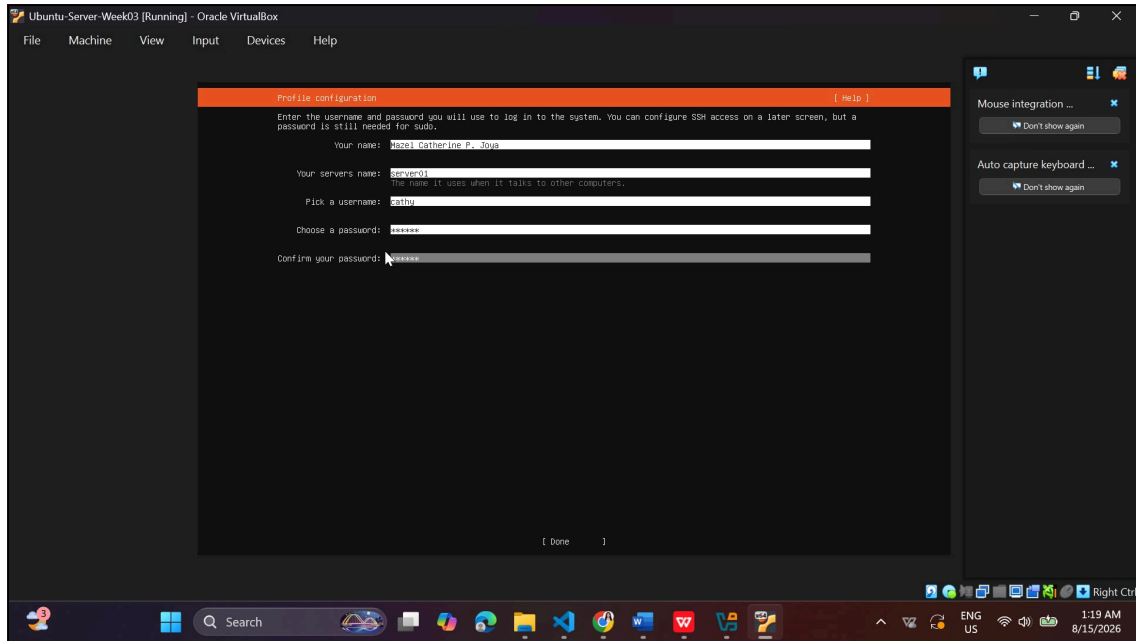


Figure 2. Profile configuration with user account cathy and hostname server01

Step 4 - Disk Partitioning

Selected **Guided - Use an Entire Disk** with the **LVM group** option enabled for flexible storage management on the 40 GB virtual disk. The default partition scheme created by the installer uses an EFI System Partition, a boot partition, and an LVM volume group for the root filesystem.

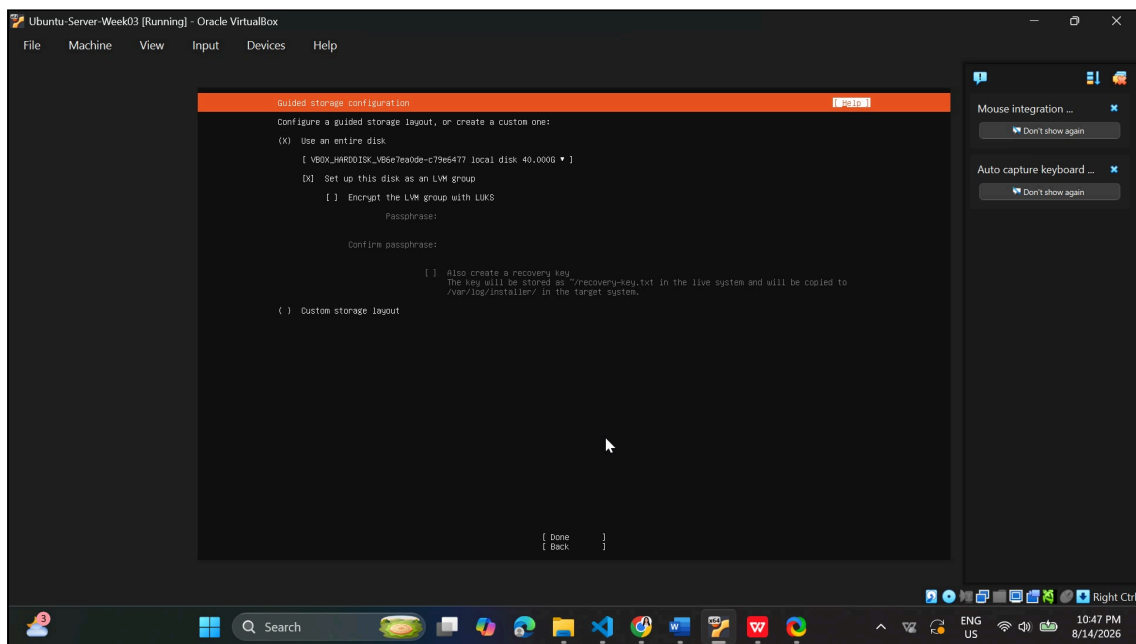


Figure 3. Guided storage configuration using the entire disk with LVM

Step 5 - SSH Server

Enabled **Install OpenSSH Server** so the server can be administered remotely and securely. No additional packages were installed, per module instructions.

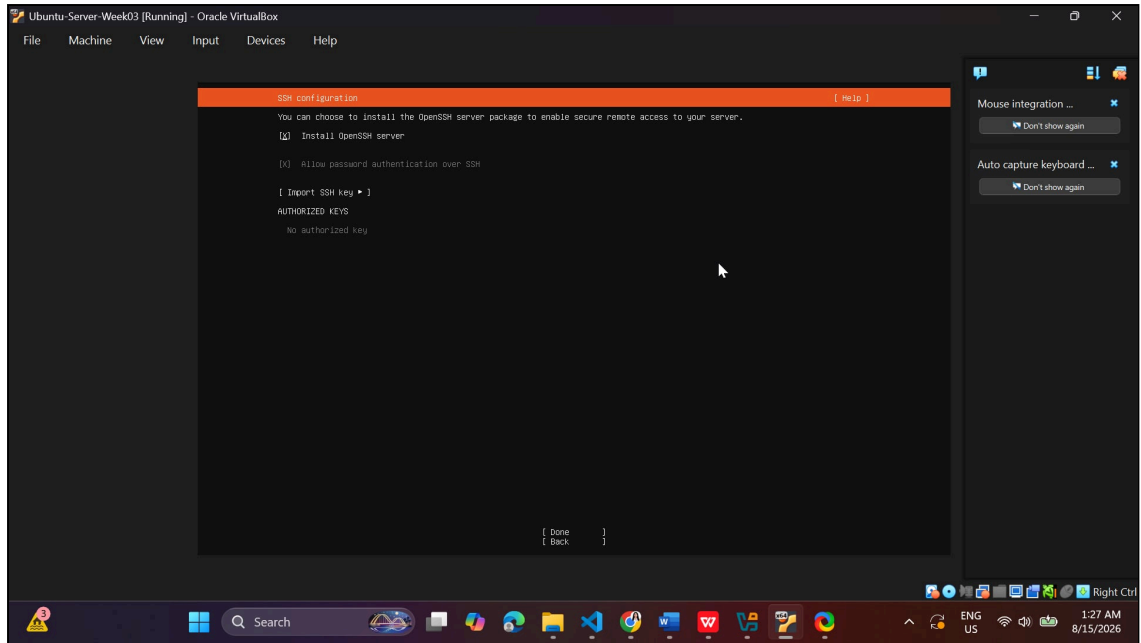


Figure 4. SSH configuration with OpenSSH server checked

Step 6 - Complete Installation

The installation proceeded successfully. The system restarted after completion and the installation ISO was removed as prompted.

6. Configuration Details

Setting	Configured Value	Purpose
Hostname	server01	Identifies the server on the network
Username	cathy	Non-root administrative account
Network	DHCP - 10.0.2.15/24 (enp0s3)	Automatic addressing on the NAT network
Disk	40 GB, Guided - Entire Disk with LVM	Default enterprise partition scheme
SSH	OpenSSH Server installed	Secure remote administration
Packages	None additional	Followed module instructions

7. Server Verification Procedures (Part B).

After installation, the server was verified using the following tasks. Each task was executed on the terminal and the result was captured.

Task 1 - Login

Logged in using the account `cathy` created during installation. The system displayed the Ubuntu 26.04 LTS message of the day.

[illegible]

Figure 5. Successful login to the server

Task 2 - Verify Hostname

```
cathy@server01:~$ hostname  
server01
```

```
cathy@server01:~$ hostname
server01
cathy@server01:~$ _
```

Figure 6. Hostname verification returning server01

Task 3 - Verify IP Address

```
cathy@server01:~$ ip addr
```

The `enp0s3` interface shows the assigned IP address **10.0.2.15/24**, matching the address recorded during installation.

```
cathy@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:b0:76:06 brd ff:ff:ff:ff:ff:ff
    altname enx080027b07606
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85849sec preferred_lft 85849sec
    inet6 fd17:625c:f037:2:a00:27ff:feb0:7606/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86035sec preferred_lft 14035sec
    inet6 fe80::a00:27ff:feb0:7606/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
cathy@server01:~$ _
```

Figure 7. IP address verification via ip addr

Task 4 - Test Internet Connectivity

```
cathy@server01:~$ ping -c 4 google.com
```

Result: **4 packets transmitted, 4 received, 0% packet loss** with latency between 9-15 ms. Internet connectivity is confirmed.

```
cathy@server01:~$ ping -c 4 google.com
PING google.com (142.250.207.46) 56(84) bytes of data:
64 bytes from ncmnlh-ai-in-f14.1e100.net (142.250.207.46): icmp_seq=1 ttl=255 time=15.9 ms
64 bytes from ncmnlh-ai-in-f14.1e100.net (142.250.207.46): icmp_seq=2 ttl=255 time=9.60 ms
64 bytes from ncmnlh-ai-in-f14.1e100.net (142.250.207.46): icmp_seq=3 ttl=255 time=9.61 ms
64 bytes from ncmnlh-ai-in-f14.1e100.net (142.250.207.46): icmp_seq=4 ttl=255 time=12.5 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 9.597/11.889/15.880/2.583 ms
cathy@server01:~$
```

Figure 8. Ping test to google.com with 0% packet loss

Task 5 - Update the Server

```
cathy@server01:~$ sudo apt update
cathy@server01:~$ sudo apt upgrade -y
```

Result: **53 packages** were upgradable; **48 packages** were upgraded successfully.

```
cathy@server01:~$ sudo apt update
Hit:1 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
53 packages can be upgraded. Run 'apt list --upgradable' to see them.
cathy@server01:~$
```

Figure 9. apt update refreshing package lists

```

cathy@server01:~$ sudo apt update
Hit:1 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
53 packages can be upgraded. Run 'apt list --upgradable' to see them.
cathy@server01:~$ sudo apt upgrade -y
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14      linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14      pollinate
Use 'sudo apt autoremove' to remove them.

Upgrading:
  apparmor      libapparmor1      linux-firmware-marvell-wireless  packagekit
  apport      libfuwud3      linux-firmware-mediatek      python3-apport
  apport-core-dump-handler  libgcrp120      linux-firmware-mellanox-spectrum  python3-distupgrade
  base-files      libpackagekit-glib2-1.0      linux-firmware-misc      python3-problem-report
  bind9-dnswtills  libsgutils2-1.48      linux-firmware-netronome      sg3-utils
  bind9-host      linux-firmware-amd-graphics      linux-firmware-nvidia-graphics  sg3-utils-udev
  bind9-libs      linux-firmware-amd-misc      linux-firmware-qlogic      ubuntu-kernel-accessories
  cloud-init      linux-firmware-broadcom-wireless  linux-firmware-qualcomm-graphics  ubuntu-minimal
  cloud-init-base  linux-firmware-intel-graphics      linux-firmware-qualcomm-misc      ubuntu-release-upgrader-core
  fwupd      linux-firmware-intel-misc      linux-firmware-qualcomm-wireless  ubuntu-server
  glr1.2-packagekitglib-1.0  linux-firmware-intel-wireless      linux-firmware-realtek      ubuntu-server-minimal
  iproute2      linux-firmware-marvell-presteria  motd-news-config      ubuntu-standard

Not upgrading yet due to phasing:
  grub-pc  grub-pc-bin  grub2-common  python3-software-properties  software-properties-common

Summary:
  Upgrading: 48, Installing: 0, Removing: 0, Not Upgrading: 5
  Download size: 584 kB / 713 MB
  Space needed: 1.198 kB / 10.6 GB available

```

Figure 10. apt upgrade installing 48 updated packages

```

Preparing to unpack ./48-packagekit-1.3.4-3ubuntu1.1amd64.deb...
Unpacking packagekit (1.3.4-3ubuntu1.1) over (1.3.4-3ubuntu1)...
Setting up motd-news-config (14ubuntu6.1)...
Setting up ubuntu-kernel-accessories (1.570.2)...
Setting up libapparmor1:amd64 (5.0.2-0ubuntu1~26.04.1)...
Setting up linux-firmware-broadcom-wireless (20260319.git217ca6e4-0ubuntu1.1)...
Setting up bind9-libs:amd64 (1:9.20.24-1ubuntu0.1)...
Setting up python3-problem-report (2.34.1-0ubuntu0.1)...
Setting up iproute2 (6.19.0-1ubuntu1.1)...
Setting up linux-firmware-nvidia-graphics (20260319.git217ca6e4-0ubuntu1.1)...
Setting up linux-firmware-mellanox-spectrum (20260319.git217ca6e4-0ubuntu1.1)...
Setting up libpackagekit-glib2-1.0:amd64 (1.3.4-3ubuntu1.1)...
Setting up linux-firmware-qlogic (20260319.git217ca6e4-0ubuntu1.1)...
Setting up libsgutils2-1.48:amd64 (1.48-3ubuntu3.1)...
Setting up linux-firmware-amd-graphics (20260319.git217ca6e4-0ubuntu3.1)...
Setting up libfwupd3:amd64 (2.1.1-1ubuntu3.1)...
Setting up python3-distupgrade (1:26.04.22)...
Setting up cloud-init-base (26.1-0ubuntu3~26.04.1)...
Encountered debconf setting for cloud-init-base/datasources.
Setting up python3-apport (2.34.1-0ubuntu0.1)...
Setting up apparmor (5.0.2-0ubuntu1~26.04.1)...
Installing new version of config file /etc/apparmor.d/Xorg...
Installing new version of config file /etc/apparmor.d/abstractions/audio...
Installing new version of config file /etc/apparmor.d/abstractions/authentication...
Installing new version of config file /etc/apparmor.d/abstractions/base...
Installing new version of config file /etc/apparmor.d/abstractions/dovecot-common...
Installing new version of config file /etc/apparmor.d/abstractions/fonts...
Installing new version of config file /etc/apparmor.d/abstractions/nameservice...
Installing new version of config file /etc/apparmor.d/abstractions/nameservice-strict...
Installing new version of config file /etc/apparmor.d/abstractions/postfix-common...
Installing new version of config file /etc/apparmor.d/abstractions/transmission-common...
Installing new version of config file /etc/apparmor.d/abstractions/wtmp...
Installing new version of config file /etc/apparmor.d/alsamixer...
Installing new version of config file /etc/apparmor.d/brave...
Installing new version of config file /etc/apparmor.d/dig...
Installing new version of config file /etc/apparmor.d/fusemount3...
Installing new version of config file /etc/apparmor.d/irssi...
Installing new version of config file /etc/apparmor.d/linux-boot-prober...
Installing new version of config file /etc/apparmor.d/lshbk...
Installing new version of config file /etc/apparmor.d/lsub...
Installing new version of config file /etc/apparmor.d/os-prober...
Installing new version of config file /etc/apparmor.d/plasmashell...
Installing new version of config file /etc/apparmor.d/proftpd...
Installing new version of config file /etc/apparmor.d/systemd-detect-virt...
Installing new version of config file /etc/apparmor.d/tftp...
Installing new version of config file /etc/apparmor.d/transmission...
Installing new version of config file /etc/apparmor.d/tunables/coreutils...
Installing new version of config file /etc/apparmor.d/who...
Progress: [ 70%]

```

Figure 11. apt upgrade progress at 70%

Task 6 - Verify SSH Service

```
cathy@server01:~$ systemctl status ssh
```

Result: Service is **active (running)** and listening on port 22.

```
cathy@server01:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Fri 2026-08-14 18:04:57 UTC; 1h 19min ago
     Invocation: 12fbc803c34a40d8878d2cf67061a74a
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 1294 (sshd)
      Tasks: 1 (limit: 1688)
     Memory: 1.6M (peak: 2.3M)
        CPU: 102ms
     CGroup: /system.slice/ssh.service
            └─1294 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 14 18:04:56 server01 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 14 18:04:57 server01 sshd[1294]: Server listening on 0.0.0.0 port 22.
Aug 14 18:04:57 server01 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Aug 14 18:04:57 server01 sshd[1294]: Server listening on :: port 22.
cathy@server01:~$
```

Figure 12. SSH service status showing active (running)

8. BIOS vs UEFI Comparison

Aspect	BIOS	UEFI
Definition	Legacy firmware from the 1970s that lives on a motherboard chip and prepares hardware before the OS loads.	Modern firmware standard managed by the UEFI Forum, with its own drivers and a richer interface.
Boot Process	Reads the first 512 bytes of the disk (MBR) and executes whatever boot code is there.	Reads boot entries from the EFI System Partition and loads boot managers like GRUB directly.
Maximum Disk	2 TB	9.4 ZB
Partition Style	MBR, max 4 primary partitions	GPT, up to 128 partitions
Security	None, executes boot code unchecked	Secure Boot verifies signatures before execution
Speed	Slower, sequential initialization	Faster, parallel device initialization
Modern Usage	Retired, only legacy compatibility mode	Standard on all modern computers and servers

Why UEFI replaced BIOS: modern drives exceed BIOS's 2 TB limit, BIOS has zero protection against bootkits while UEFI's Secure Boot verifies every boot loader, and UEFI boots faster with parallel hardware initialization. The industry sealed the transition when Microsoft required UEFI for Windows 8 certification in 2012. The virtual machine deployed in this project boots through UEFI, which is now the standard for enterprise servers.

Full report: [BIOS_vs_UEFI.pdf](#)

9. Boot Process Flowchart

The Ubuntu boot process in 8 stages:

1. Power On
2. BIOS/UEFI Initialization (POST)
3. Boot Device Detection
4. Boot Loader (GRUB)
5. Linux Kernel
6. init/systemd (PID 1)
7. Services Start
8. Login Prompt

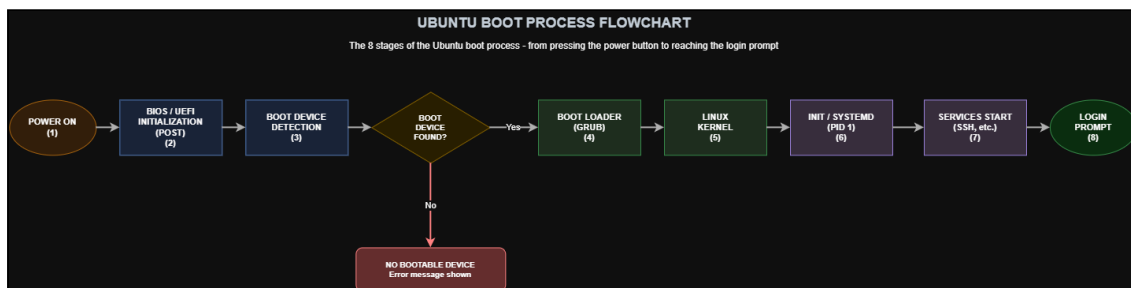


Figure 13. Ubuntu boot process flowchart (horizontal layout)

10. Windows Server Installation Summary

As required by the bring-home assignment, I also installed **Windows Server 2025 Standard Evaluation (Desktop Experience)** in a separate virtual machine.

Component	Value
VM Name	WindowsServer-Week03
Edition	Windows Server 2025 Standard Evaluation (Desktop Experience)
RAM / CPU	4096 MB / 2 processors
Storage	50 GB
Computer Name	WIN-SERVER-01
Administrator Password	Set during installation
License	Evaluation, valid 180 days

Summary of steps: created the VM in VirtualBox with the evaluation ISO, selected the Desktop Experience edition, installed on Disk 0 (unallocated space), waited through the multi-reboot installation, set the Administrator password during the OOBE "Customize settings" step, unlocked with Ctrl+Alt+Delete, logged in successfully, renamed the computer to WIN-SERVER-01 via System Properties, and verified the desktop with Server Manager.

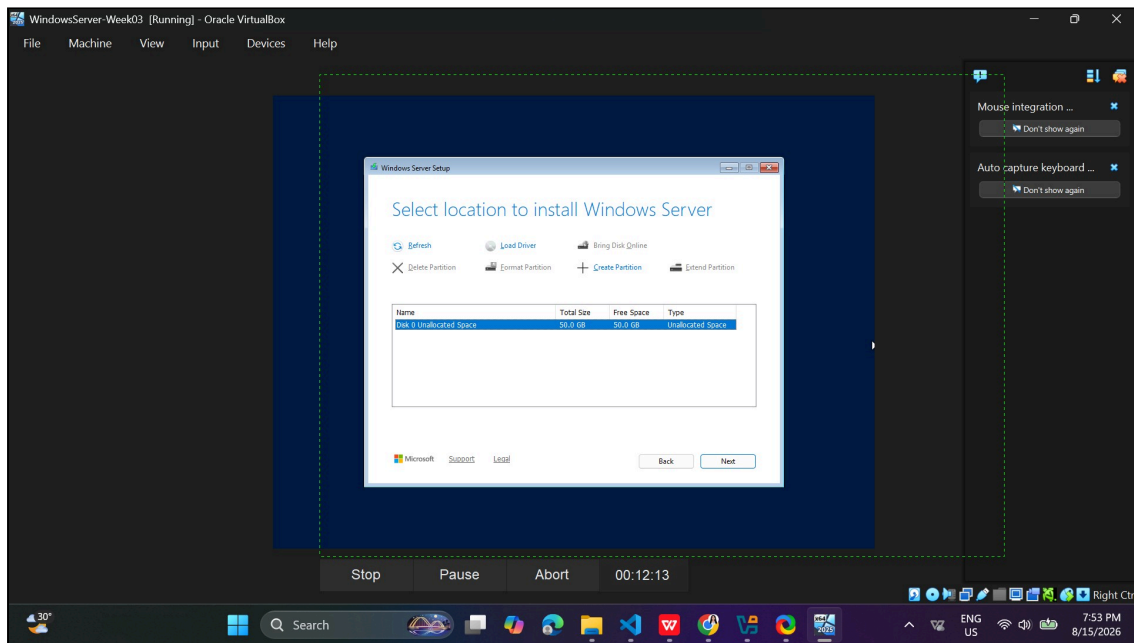


Figure 14. Selecting the installation location

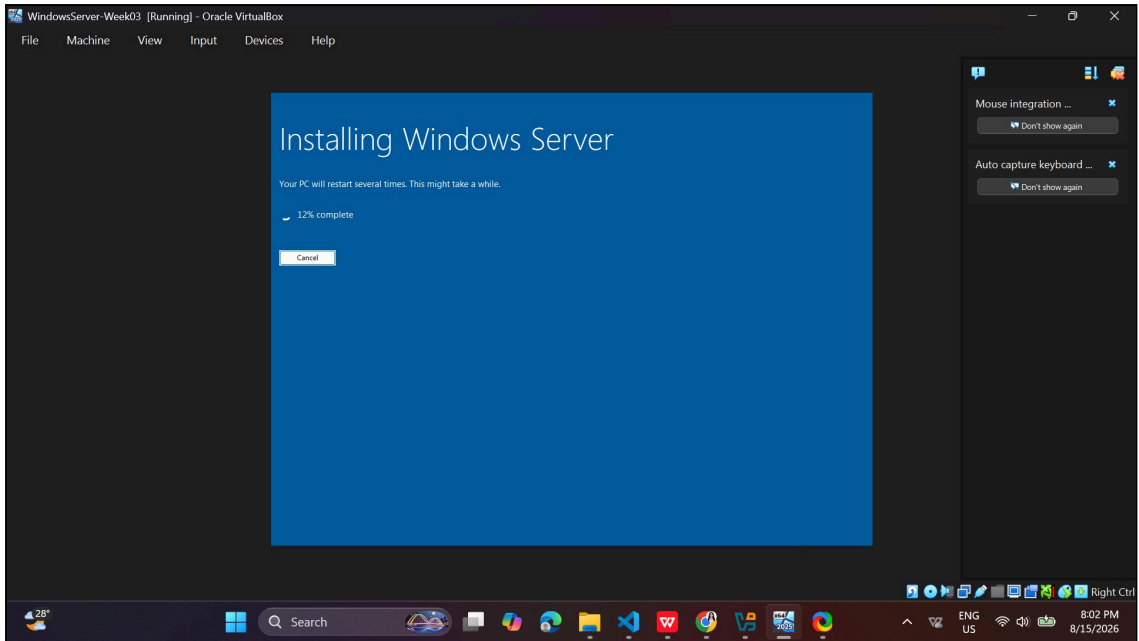


Figure 15. Installation in progress

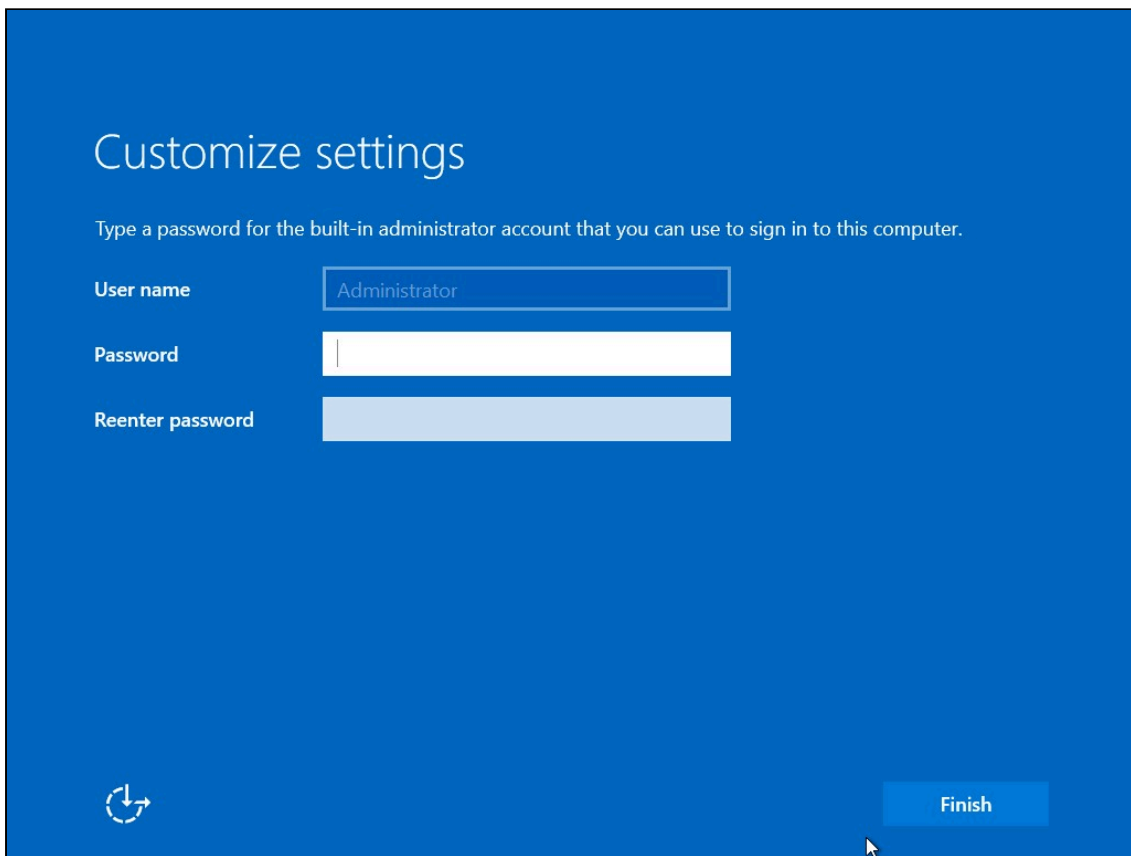


Figure 16. Setting the Administrator password

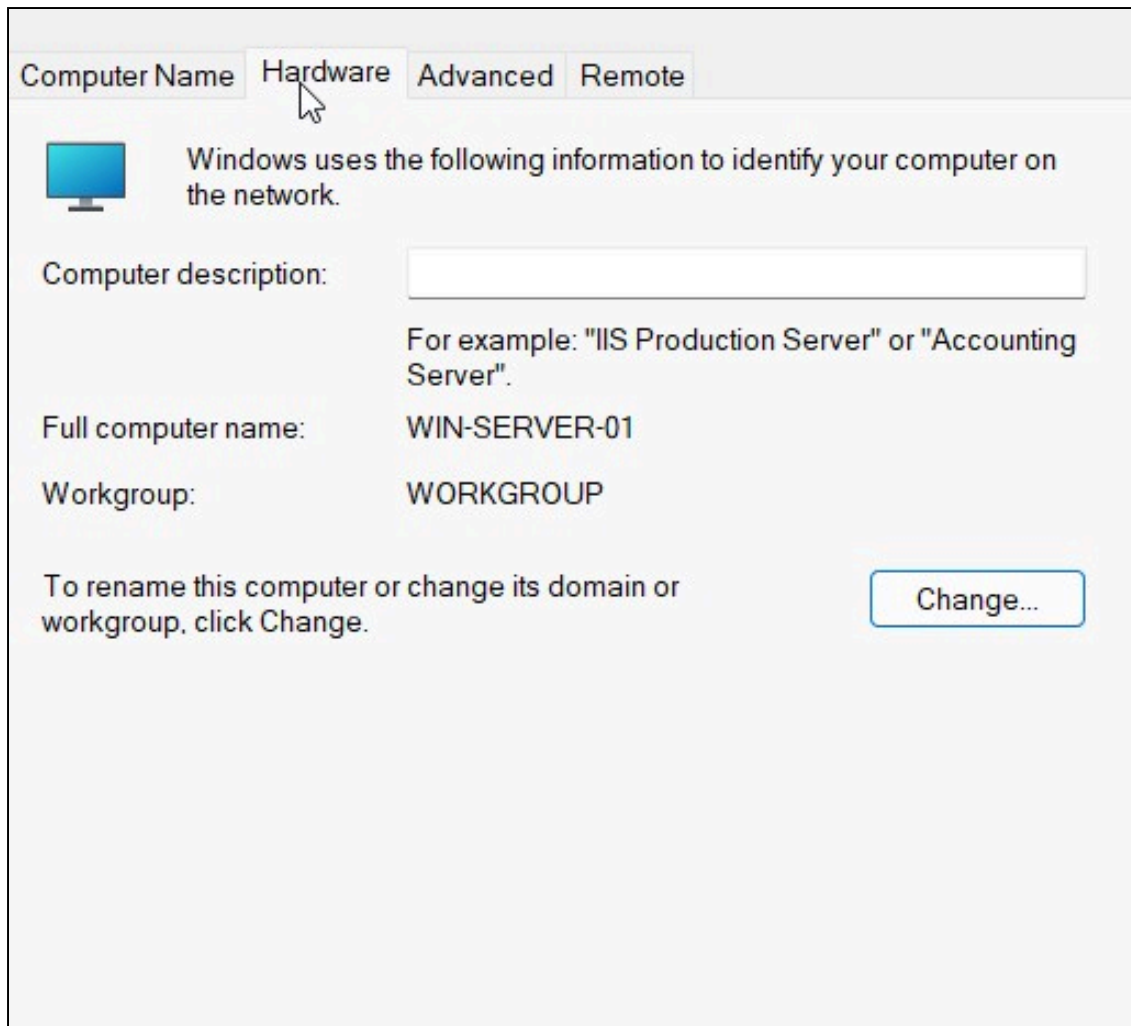


Figure 17. Computer renamed to WIN-SERVER-01

11. Operating System Comparison Table

Aspect	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Commercial, per core plus CALs	Free and open source	Free, RHEL rebuild
User Interface	GUI + PowerShell	Terminal by default	Terminal by default
Package Management	MSI / WSUS	APT (.deb)	DNF (.rpm)
Security	Secure Boot, BitLocker, Defender	AppArmor, UFW	SELinux (default on)
Performance	Heavier due to GUI	Very light footprint	Very light footprint
Enterprise Use	Active Directory, Exchange, SQL Server	Web, DB, cloud, startups	RHEL-compatible production
Advantages	Great support, AD, huge ecosystem	Free, huge community, LTS support	RHEL compat, SELinux, very stable
Disadvantages	Licensing cost, GUI overhead	CLI learning curve	Smaller community, older packages

For ABC Startup Solutions, Ubuntu Server was the right choice because of the zero licensing cost, lighter footprint, and massive community support. A company with strict compliance needs or heavy Microsoft integration would reasonably go with Rocky Linux or Windows Server instead. Full discussion: [OS_Comparison.pdf](#)

12. Problems Encountered and Solutions

#	Problem	Solution Implemented
1	The whole installation took hours. There were multiple restarts and long progress bars, and I kept waiting for each phase to finish before I could move on.	I stayed patient and let the installation complete without interrupting it. I used the waiting time to review the next steps in the module so I was ready when each screen appeared.
2	My laptop started lagging while the virtual machine was running. Every click took a few seconds and even typing felt delayed because the VM was using most of the memory.	I closed the programs I was not using, freed up space, and gave the virtual machine room to work. It taught me that managing the resources of the host machine is part of the job.
3	My first login attempt returned "Login incorrect" because I mistyped the password.	I carefully re-entered the correct password and logged in successfully. This reinforced that Linux credentials are case-sensitive and must be entered exactly.
4	Some errors appeared during the setup that I did not expect, and a few steps had to be repeated before they worked.	I read the error messages, checked what each one meant, and fixed the issues one at a time until everything worked correctly. Solving them on my own was the most useful practice of the week.

13. Lessons Learned

- An operating system is only as reliable as the choices made during installation: hostnames, passwords, partitions, and services all matter.
- Verification is proof. Commands like `hostname` , `ip addr` , `ping` , and `systemctl status ssh` turn assumptions into evidence.
- Documentation is part of the job. Recording every step lets a future administrator reproduce the installation reliably.
- Firmware knowledge matters. Understanding BIOS vs UEFI helps in choosing the right boot mode and securing the boot chain.
- No single operating system is best; the right fit depends on licensing, workload, security needs, and team skills.

14. References

1. Canonical. Ubuntu Server documentation. <https://ubuntu.com/server/docs>
2. Oracle. VirtualBox User Manual. <https://www.virtualbox.org/manual>
3. OpenSSH. OpenSSH Manual Pages. <https://www.openssh.com/manual.html>
4. UEFI Forum. UEFI Specification. <https://uefi.org/specifications>
5. Microsoft Learn. UEFI firmware requirements. <https://learn.microsoft.com/en-us/windows-hardware/design/device-experiences/oem-uefi>
6. Microsoft. Windows Server 2025 Evaluation. <https://www.microsoft.com/en-us/evalcenter/evaluate-windows-server-2025>
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